

SAP-C02 - LAST-MILE MAP

★ Exam shape

75 questions | 180 min | scaled passing score 750 | single + multiple response.

★ Domain weights

D1 Organizational Complexity 26% | D2 New Solutions 29% | D3 Continuous Improvement 25% | D4 Migration & Modernization 20%.

★ Read the stem

Circle: MOST cost-effective / LEAST ops / minimal downtime / global / RTO / RPO / compliance. Eliminate technically valid answers that miss the stated priority.

★ Default architecture

Multi-AZ first; Multi-Region only when business/RTO/data-residency requires it. Prefer managed services, loose coupling, automation, observability, least privilege.

★ Decision ladder

1) hard constraints 2) availability/durability 3) security 4) performance 5) operations 6) cost.

★ Time plan

~2 min 24 sec/question. First pass $\leq$ 2 min; flag long calculations; reserve 20-25 min to review.

★ Answer smell

Manual servers, long-lived keys, public databases, NAT for AWS-service traffic, single-AZ production, custom code where a managed feature exists.

ORGANIZATIONS, IAM & GOVERNANCE

★ Organizations

OUs group accounts. SCP = maximum permissions; it never grants. Management account is not restricted by SCPs. Explicit deny wins. Use delegated administrator.

★ Identity

Workforce -> IAM Identity Center + external IdP. Workloads -> roles. Cross-account: trusted account role + trust policy + caller permission. Avoid IAM users/keys.

★ ABAC

Use tags/principal attributes for scalable permissions. Permissions boundary caps identity permissions; session policy can further reduce them.

★ Multi-account baseline

Control Tower landing zone, Account Factory, guardrails, centralized CloudTrail/Config/Security Hub/GuardDuty, log archive + security account.

★ Resource sharing

RAM shares subnets, TGW, Route 53 Resolver rules, licenses, etc. Private Marketplace controls approved products.

★ Secrets & encryption

Secrets Manager = rotation. Parameter Store = config/secrets, cheaper. KMS key policy is essential. Cross-account encrypted use needs key policy + IAM + grants where applicable.

★ Audit

Org trail -> central immutable S3; Object Lock for WORM. Config = resource configuration/compliance. CloudTrail = API activity. Access Analyzer = unintended external access.

★ Trap

Service-linked roles are created/used by AWS services; do not replace them casually with custom roles.

NETWORKING - HYBRID & GLOBAL

★ VPC connectivity

Many VPCs -> Transit Gateway. Same-region private service exposure -> PrivateLink. VPC peering is non-transitive and has no overlapping CIDRs.

★ Hybrid

Direct Connect = private, consistent; not encrypted by default. Add VPN for encryption. DX Gateway connects DX to multiple VPCs/TGWs. Site-to-Site VPN = fast setup.

★ DNS

Inbound Resolver endpoint: on-prem -> Route 53. Outbound: VPC -> on-prem. Share Resolver rules with RAM. Private hosted zone needs VPC association.

★ Endpoints

Gateway endpoint: S3/DynamoDB, no hourly charge. Interface endpoint: PrivateLink/ENI + SG + hourly/data fees. Endpoint policies add control.

★ Global ingress

CloudFront = cache/HTTP edge. Global Accelerator = static anycast IP, TCP/UDP, fast regional failover. Route 53 = DNS routing; health checks/failover.

★ NAT

NAT Gateway is AZ-specific. One per AZ with same-AZ routes avoids cross-AZ dependency/cost. NAT supports outbound only; use endpoints to reduce NAT traffic.

★ Inspection

AWS Network Firewall / centralized inspection VPC. TGW appliance mode preserves symmetric routing for stateful appliances.

★ IPv6

Egress-only internet gateway gives outbound-only IPv6. NAT Gateway is primarily IPv4; plan dual-stack routes and SG/NACL rules.

STORAGE & DATA TRANSFER

★ S3

Versioning + replication for recovery. CRR/SRR needs versioning. Replication is asynchronous. Object Lock compliance mode cannot be bypassed during retention.

★ S3 tiers

Intelligent-Tiering for unknown/changing access. Standard-IA min 30 days; Glacier Instant 90; Flexible 90; Deep Archive 180. Retrieval/early-delete charges matter.

★ S3 security

Block Public Access + bucket policy + SSE-KMS. Presigned URL = temporary object access. Access Point = per-app policy. MRAP = multi-region access/routing.

★ File/block

EBS = single-AZ block (multi-attach limited). EFS = regional Linux NFS. FSx Windows = SMB/AD. FSx Lustre = HPC/S3. ONTAP = multiprotocol/snapshots.

★ Transfer

DataSync = online file/object transfer + validation. Storage Gateway = hybrid access. Snow Family = offline/edge when bandwidth/time is limiting. Transfer Family = SFTP/FTPS/FTP.

★ Backup

AWS Backup centralizes policies/cross-account/region copies; Vault Lock for WORM. EBS snapshots incremental; Fast Snapshot Restore removes initialization latency.

★ Large data question

Calculate transfer time and egress. If network misses deadline, choose Snow. For recurring online sync, choose DataSync.

★ Database migration

DMS moves data/minimal downtime (full load + CDC); SCT/schema conversion or DMS Schema Conversion for heterogeneous engines.

COMPUTE, CONTAINERS & SERVERLESS

★ EC2 purchasing

On-Demand = variable. Savings Plans = steady compute flexibility. Reserved Instances = EC2/RDS reservation benefits. Spot = fault-tolerant; interruption handling.

★ Auto Scaling

Target tracking simplest. Step scaling for metric bands. Predictive + dynamic for recurring demand. Warm pools reduce startup. Instance refresh rolls AMI changes.

★ Load balancers

ALB = HTTP/L7/path/host/Lambda. NLB = TCP/UDP/TLS/static IP/high performance. GWLB = virtual appliances. Cross-zone behavior/cost can matter.

★ ECS/EKS

Fargate = no server management. ECS simpler AWS-native orchestration. EKS when Kubernetes ecosystem/portability required. Use ECR scanning/signing and IAM roles for tasks/service accounts.

★ Lambda

Event-driven short tasks; concurrency, retries and idempotency matter. Provisioned concurrency for predictable cold-start latency. VPC access for private resources.

★ Queues/events

SQS buffers work; visibility timeout > processing time. FIFO for ordering/deduplication. SNS fanout. EventBridge routing/SaaS/schedules. Step Functions orchestration.

★ API

API Gateway = managed APIs/auth/throttling/caching. SQS integration decouples spikes. WAF protects common web exploits.

★ Images

Golden AMI via EC2 Image Builder + pipeline. Systems Manager Automation/Patch Manager for fleet operations; no inbound SSH/RDP needed with Session Manager.

DATABASE DECISIONS

★ **RDS/Aurora**

RDS Multi-AZ = HA, synchronous standby, not read scaling. Read replica = read scaling/DR, async. Aurora replicas provide reads + fast failover.

★ **Aurora global**

Global Database = low-latency global reads and cross-region DR; storage replication. Know planned switchover vs unplanned failover and possible data loss.

★ **DynamoDB**

Key-value, massive scale. On-demand for unpredictable traffic; provisioned + auto scaling for steady. Global Tables = active-active multi-region. DAX = microsecond read cache.

★ **Caching**

ElastiCache Redis/Valkey = rich data structures, persistence, HA. Memcached = simple sharded cache. Do not use cache as system of record.

★ **Specialized**

Redshift = warehouse/OLAP. OpenSearch = search/log analytics. Neptune = graph. Timestream = time series. DocumentDB = Mongo-compatible workloads.

★ **RPO/RTO**

Automated backups/PITR protect data; replicas improve availability. Cross-region replica/copy protects regional event. Match technology to explicit RPO/RTO.

★ **Migration**

Homogeneous: native tools or DMS. Heterogeneous: schema conversion + DMS. Babelfish may ease SQL Server app migration to Aurora PostgreSQL.

★ **Consistency trap**

DynamoDB strongly consistent reads are same-region only and cost more RCUs; GSIs and global replication are eventually consistent.

RELIABILITY, DR & OPERATIONS

★ DR ladder

Backup/restore: cheapest, longest RTO. Pilot light: core always on. Warm standby: scaled-down functional stack. Multi-site active-active: lowest RTO, highest cost/complexity.

★ Resilience

Spread across AZs; remove single points; use health checks and self-healing. Cell-based isolation limits blast radius. Static stability keeps dependencies available during failure.

★ CloudFormation

StackSets = multi-account/region. Change sets preview. Drift detection finds manual change. Nested stacks reuse patterns. DeletionPolicy Retain/Snapshot protects data.

★ Monitoring

CloudWatch metrics/logs/alarms; Logs Insights queries. X-Ray traces. CloudTrail API audit. Config state/compliance. EventBridge reacts to state/API events.

★ Ops automation

Systems Manager: inventory, patching, Run Command, Automation, Session Manager, State Manager. OpsCenter/AWS Incident Detection and Response where appropriate.

★ Deployment

Blue/green minimizes risk; canary shifts small traffic; rolling replaces batches. CodeDeploy supports traffic shifting/rollback. Database change must be backward compatible.

★ Quotas

Service Quotas + monitoring before migrations/events. Scale tests, game days, runbooks, and dependency mapping are part of architecture.

★ Well-Architected

Security, Reliability, Performance Efficiency, Cost Optimization, Operational Excellence, Sustainability. Questions often trade among pillars.

SECURITY - FAST ELIMINATION RULES

★ DDoS/WAF

Shield Standard automatic. Shield Advanced adds stronger protection, cost protection and DRT. WAF = L7 rules/rate limits/bot controls. Firewall Manager = org-wide policy.

★ Threat findings

GuardDuty = threat detection. Security Hub = aggregate/posture. Inspector = EC2/ECR/Lambda vulnerability. Macie = S3 sensitive data. Detective = investigation.

★ Private access

Keep databases private. Use VPC endpoints, private APIs, CloudFront OAC for S3 origins. Prefer SSM Session Manager over bastions.

★ Encryption

At rest: KMS. In transit: TLS/ACM. CloudHSM when dedicated HSM/control is required. Imported key material/external key store only when requirement justifies ops burden.

★ Certificates

ACM public certs integrate with AWS services and auto-renew when validation/association conditions hold. Private CA for private PKI. CloudFront cert must be in us-east-1.

★ Data perimeter

Use aws:PrincipalOrgID, source VPC/VPCE, resource/account conditions. Beware confused deputy: aws:SourceArn and aws:SourceAccount.

★ Incident

Contain first: isolate SG, revoke sessions/keys, preserve snapshots/logs. Automate response with EventBridge + Lambda/SSM. Keep evidence in immutable central account.

★ Shared responsibility

AWS secures cloud infrastructure; customer secures identities, data, configuration, guest OS (EC2). Managed service shifts more work to AWS.

COST, MIGRATION & EXAM TRAPS

★ **Cost tools**

Cost Explorer analyzes. Budgets alerts/actions. CUR = most detailed billing data. Cost Anomaly Detection spots unusual spend. Compute Optimizer rightsizes.

★ **Cost levers**

Rightsize -> schedule nonprod -> Graviton -> Savings Plans/RIs -> Spot for tolerant work -> storage lifecycle -> endpoints/CloudFront -> reduce cross-AZ/region/NAT data.

★ **Migration phases**

Assess portfolio -> mobilize/landing zone -> migrate waves -> validate/optimize. Application Discovery Service / Migration Evaluator support discovery/business case.

★ **7 Rs**

Retire, Retain, Rehost, Relocate, Replatform, Repurchase, Refactor. VMware move with minimal change may be Relocate; lift-and-shift servers = Rehost.

★ **Tools**

Application Migration Service = block-level server migration. DMS = databases. DataSync = files/objects. Migration Hub = tracking. Elastic Disaster Recovery = ongoing DR replication.

★ **Common traps**

Multi-AZ != read scaling. Read replica != zero-RPO HA. S3 replication != backup alone. SCP != permission grant. DX != encrypted. SG stateful; NACL stateless.

★ **When two answers work**

Choose the one satisfying every hard constraint with fewer components and less custom operations. AWS exams reward managed, automated, scalable patterns.

★ **Final check**

For multi-select, verify every selected option is independently required. Do not add a “nice to have.” Re-read qualifiers and confirm the architecture end-to-end.